Distributions, and the Summaries That Hide Them

Four hundred House districts drawn four ways, and an average that describes almost nothing

Democracy’s Data

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The American House district, you will read, is a fifty-fifty proposition: the average district splits its vote almost evenly between the parties. The average is real. In the 2024 election, the Democratic share of the two-party vote across the 397 contested races had a mean of **49.8%** and a median of **49.6%**. The two agree to within 0.20 of a point. So is the typical district a coin flip?

The question matters because summaries are how citizens meet data. Nobody hands you 397 numbers; they hand you one, and the one number decides what you believe about whether your own vote is contested. This brief draws the same column four ways: a box, a histogram, a cumulative curve, and a stack of years. The same mean, it turns out, can sit on top of opposite electorates.

01 The test

The data is the Democratic share of the two-party vote in every contested U.S. House race from 1946 to 2024. The series joins Gary Jacobson’s House elections file to the Clerk of the House’s *Statistics of the Congressional Election* for the recent years. Both are read through this book’s house-competition chapter, which assembles them and documents its repairs; the figures here read that assembled file rather than rebuilding it.

The question needs this source because it is about every district at once. A summary hides its distribution only when you hold the whole distribution. House returns are the one record with a number for every district in every election: complete and official rather than sampled, for the reasons this cluster’s chapter on election returns lays out. The test itself is simple: compute the standard summaries, then draw the actual 397 numbers several ways and see which claims survive.

02 The data

One row of `districts.csv` is one House race in one election year. A second table, `by_year.csv`, carries one row per election with the summaries already computed. The columns of the district file, by name:

Column	What it holds	Measurement
year	the election year	discrete, 1946 to 2024
stcd	a state-district code — the race's identifier	categorical, one per race
dv	the Democratic share of the two-party vote, 0 to 100	continuous
uncontested	whether only one candidate stood	dichotomous

`dv` is a **share of the two-party vote**, not a share of all votes and not a margin. A district at 50 split evenly between the two major parties; a district at 70 gave the Democrat seven of every ten major-party votes. Four rows of 2024, from the two ends of the column:

District code	Democratic share	What it means
2004	88.7%	the most Democratic contested seat
1005	85.7%	the second most
4201	19.9%	the second least
2703	19.6%	the least Democratic contested seat

The cleaning decision that matters most is an absence. 38 of the 435 seats in 2024 had only one major-party candidate, so there is no second party and no share to compute. In those rows `dv` is missing, and those seats are not among the 397 drawn below. They are also, by definition, the least competitive seats in the country. Every figure in this brief is therefore drawn on a set with the extreme already removed, a deletion that is patterned rather than random — the ordinary condition of a computed variable.

Here are the summaries of the 397 that remain:

Summary	Value
Mean	49.8%
Median	49.6%
Standard deviation	15.7
25th percentile	36.4%
75th percentile	62.0%
Within 5 points of even	16.9% of races
Within 10 points of even	37.3% of races

Read that table on its own and a picture assembles itself: a distribution centred a shade under even, spread about fifteen points either way, with half the districts between 36.4 and 62.0. It is the picture a well-behaved bell would produce, and every number in the table is correct.

> **Before you read on, commit to a number.** On the evidence above, **how many of the 397 districts do you expect to sit within five points of even?** Write it down. The table gives you everything you need to guess, and the guess is the point.

03 What it says about democracy

A box says one thing. Start with the form the summary table implies.

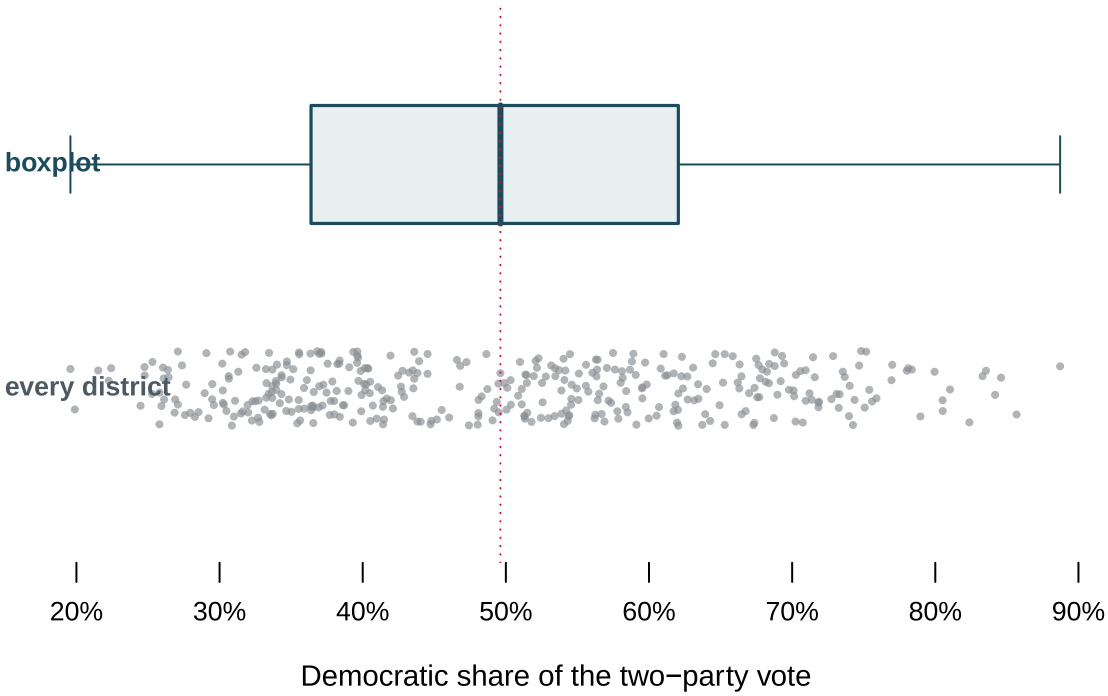

Figure 1. The same 397 numbers, twice. A boxplot above, and below it the districts themselves, jittered so they do not stack. This paired form fits the question because the argument is that the second row contradicts the first, and only one shared axis makes that visible. Hover any point for its district; the dashed line carries the median down from the box.

The boxplot is not wrong: every mark on it is a correct summary of the numbers below it. But the dashed line falls through a **thin patch** in the strip. The box has quietly asserted that the middle of the data is where the data is densest, and in this variable that is false.

A histogram says another.

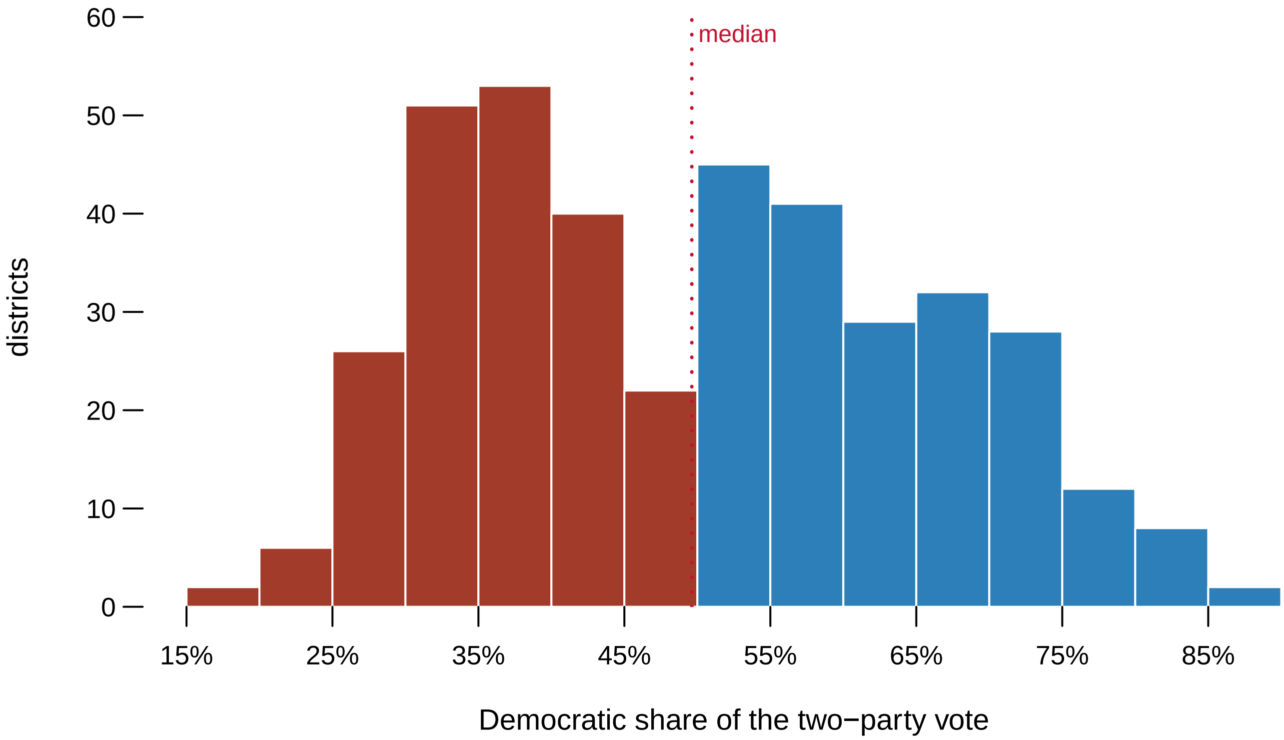

Figure 2. The same numbers again, binned. The control sets the bin width; nothing else changes. At five points the shape is unmistakable and it is not a bell — it is **two hills with a valley between them**, and the median sits in the valley. There are 22 districts in the five points below even, against 40 in the five points below that and 45 in the five points above.

Now widen the bins to twelve or fifteen. The valley closes, the hills merge, and the picture becomes the single well-behaved hill the summary table implied. **No number changed.** The only thing that changed is a parameter nobody is obliged to report, and a histogram published without its bin width is an argument with a step missing. The two peaks sit near 35.6% and 55.5%, and a district at either peak is a seat one party wins comfortably. The modal American House district is safe, and there are two kinds of it.

A cumulative curve answers the question you asked. A histogram is a good way to see a shape and a bad way to answer *how many*. The opening question, how many districts are close to even, is a counting question, and there is a form that answers it exactly.

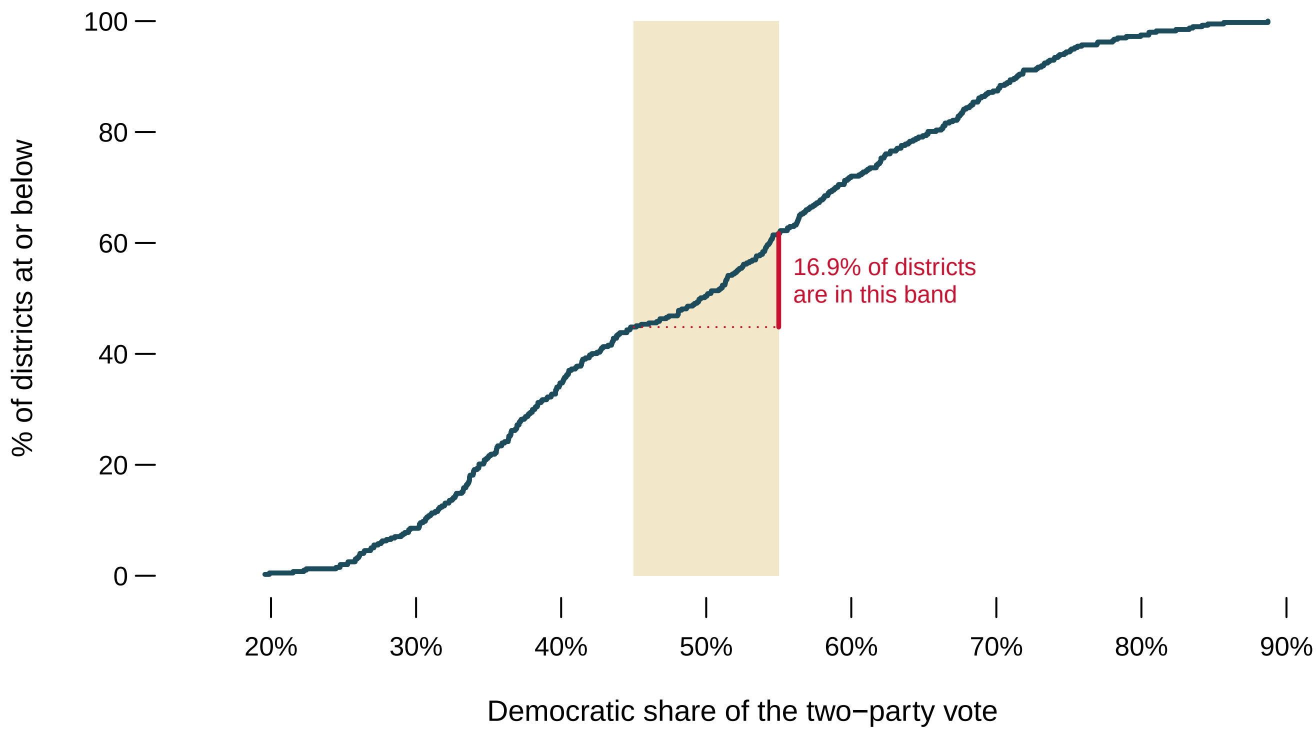

Figure 3. The same numbers as a cumulative curve. For any share on the horizontal axis, the curve gives the percentage of districts at or below it. Set the band and the figure reads the count off the curve directly: at ±5 points it is **16.9%** of contested districts, and at ±10 points **37.3%**.

Compare that with the number you wrote down. The summary table said the middle half of districts ran from 36.4 to 62.0, a spread of 25.7 points, and it is easy to read that as *the middle is crowded*. It is not. The curve climbs steeply on both flanks and **flattens through the middle**, and a flat stretch on a cumulative curve is exactly what an empty region looks like.

Five elections, stacked. The valley is often explained as something recent, a product of the last two redistricting cycles. One year cannot test that; a stack of years can.

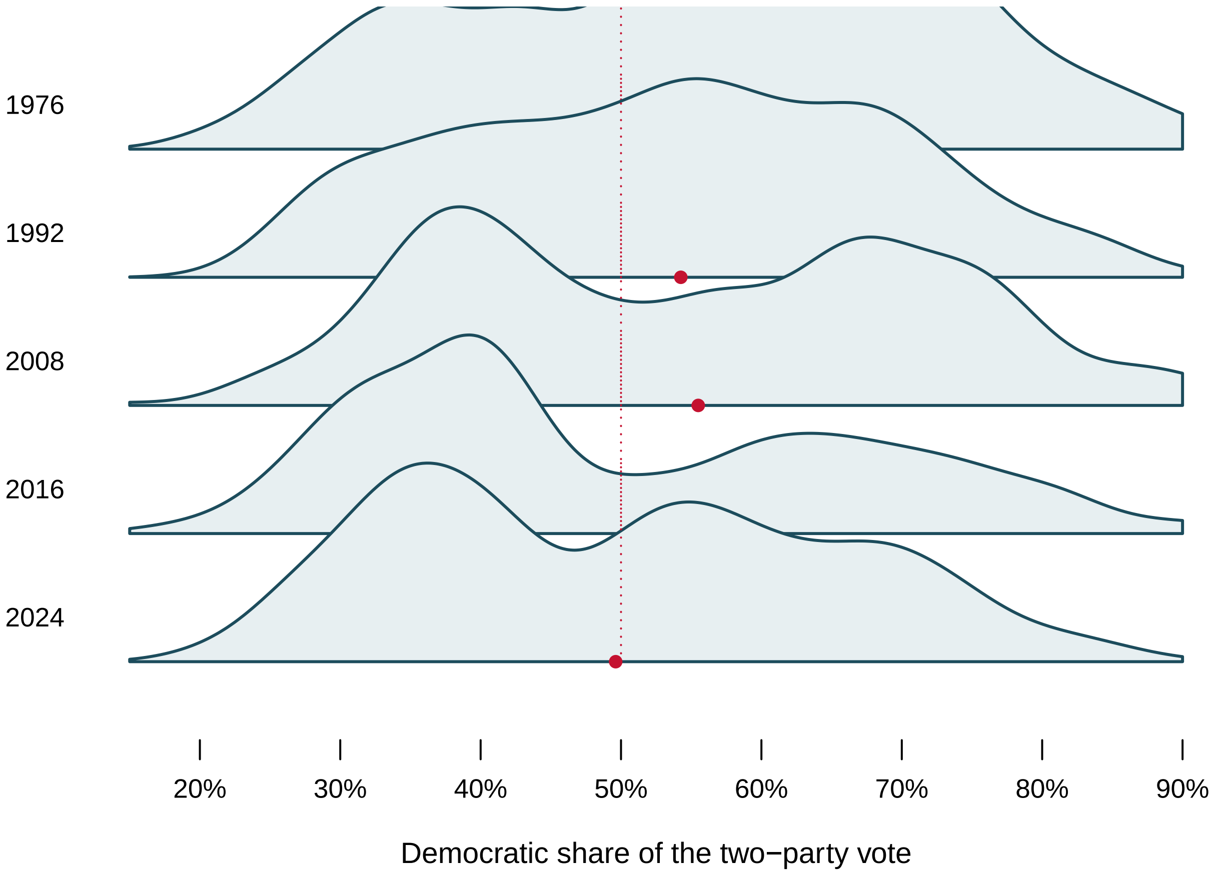

Figure 4. Five elections, same bandwidth, same axis. The dashed line is an even split and the red dot is that year's median. Hover a band for its counts.

The valley is not new. It is there in 1976, when only 17.2% of contested races fell within five points of even — a *lower* share than in 2024. What has moved is not the hole but the position of the whole distribution: in the 1970s and 1980s it sat well to the Democratic side of even, and it has slid since. A reader shown only 2024 would reasonably conclude the two-hill shape is a modern artifact of redistricting. Five elections say otherwise, and the only reason to believe the stack over the single year is that it contains more of the record.

04 What this brief cannot tell you

It cannot show you the least competitive seats at all. The 38 uncontested seats of 2024 have no two-party share, so they are absent from every figure above. The valley would be no shallower with them; the flanks would be taller.

It cannot name an honest form. The boxplot's quartiles are correct. The histogram is correct at every bin width, and the density curves in Figure 4 use one bandwidth chosen once and applied everywhere, which is a defensible choice and still a choice. The argument is not that some charts deceive and others do not. It is that **every summary is a choice about what to discard**, and the only defence is to draw more than one and say which you chose.

A share is not a seat. Nothing here counts who won. A district at 49% elected a Republican and a district at 51% elected a Democrat, and this brief has treated them as neighbouring numbers, which they are, and as similar outcomes, which they are not.

05 What you should have learned

- A mean of 49.8% and a median of 49.6% agree to within 0.20 of a point, and both sit in the emptiest part of the distribution.
- The 2024 House is two hills, not a bell: peaks near 35.6% and 55.5%, with only 16.9% of contested districts within five points of even.
- A histogram's finding can appear and disappear with its bin width while no number changes, which is why the bin width has to travel with the chart.
- The valley is not a product of recent redistricting: 1976 put 17.2% of contested races within five points of even, less than 2024.
- The 38 uncontested seats of 2024 are missing from every figure, and their absence is patterned, not random.

06 Extensions

None of these is answered above, and every one can be answered from the tables the Sources section hands over.

- `by_year.csv` has `mean` and `median` for every year alongside `sd`. Find the years where mean and median are furthest apart. What is the distribution doing there?
- `by_year.csv` also counts `uncontested` seats. Follow that column across the series. When was the ballot emptiest, and does it move with the `within5` column or against it?
- `by_year.csv` carries `within5` and `within10`. Follow them from 1946 to 2024. Has the number of genuinely close districts fallen, and by how much?
- `districts.csv` is one row per district per year, with the share in `dv`. Pick a year this brief does not draw and build its histogram in a spreadsheet. Then say what that year's mean was hiding.

Two stretch ideas point past the spreadsheet. Senate races have the same two-party share and only 33-odd contests per cycle; pull one year's results from the Clerk's *Statistics* and see whether so few races can even have a valley. And find a published news chart of "average district partisanship" and work out which of the four forms above it silently chose.

07 Sources

Data

- Jacobson, Gary C. *House Elections Data, 1946–2014.* Democratic share of the two-party vote, by district and year.
- Clerk of the U.S. House of Representatives. *Statistics of the Congressional Election*, 2016 through 2024 editions. https://clerk.house.gov/Votes/ElectionStatistics

- Both are read through this book’s house-competition chapter, which assembles them and documents its repairs.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`by_year.csv`, `districts.csv`